

Course Code:	DATA308
Course Title:	Generative AI
Assignment #1, #2	Academic Decision-Support System for University
Deadlines	#1: Submission: 22 nd September 2026 EOD Presentation: 28 th September 2026 #2: Submission: 3 rd November 2026 EOD Presentation: 9 th November 2026
Deliverables	a. 10 Page Report of work with Demo Link of the system b. Presentation of the work carried out
Max Marks	50 (Will be scaled down to 25) for each assignment
Penalty	2 marks will be deducted for each day the submission is late. - Submissions received more than one week late will not be accepted.

Assignment Title: “*Design, Build, Deploy and Evaluate an AI Academic Advisor for University that knows when to answer, what to answer, when to ask and when to say it does not have enough information.*”

Assignment #1

1. Assignment Question

Design and evaluate an AI Academic Advisor that can answer student academic queries using a given set of university academic regulations, programme/course structure, course prerequisites, semester-wise course offerings, and other academic documents. The objective is not simply to upload the documents and ask questions, but to investigate how prompt engineering, structured information, RAG and optionally agentic workflows can improve the accuracy, reliability and usefulness of academic advice. Since no personal student database will be provided, you are required to create synthetic/anonymized student profiles themselves for testing. The system should be able to answer academic questions, identify when information is insufficient or conflicting, provide supporting sources where possible and avoid making unsupported recommendations. Compare your approach against a basic LLM/document-Q&A baseline and demonstrate the strengths and limitations of their solution.

You are encouraged to be creative in designing the system.

2. Data Provided

The Data provided includes university academic information package, which will include:

- Academic Regulations — academic policies, credit requirements, graduation requirements, attendance, progression rules, etc. (Student Handbook)
- Programme Structure — core courses, minors, specializations, etc.
- Course Catalogue — course codes, titles, credits, descriptions, prerequisites/corequisites.
- Semester-wise Course Offering — courses offered in different semesters.
- SOP for the semester

3. Constraints

- No real student personal information will be provided.
- Create synthetic student profiles in a structured format (e.g., JSON/CSV/table) and use them to test student-specific queries.
 - Student A: Completed 72 credits, passed Linear Algebra and Programming in Python, failed Data Structures, etc.
 - These profiles should not contain real student names, IDs, contact details or other personal information.
- The AI Advisor should ground its academic recommendations in the provided university documents and should clearly indicate when the available information is insufficient to answer a query.
- The AI Advisor should answer only questions pertaining to the context.

4. Assignment Phases:

(BUILD → CHALLENGE → IMPROVE → MEASURE → ANALYZE → DEPLOY → EXTEND → PRESENT)

Phase #1 — Build the AI Academic Advisor

- Develop a RAG-based AI Academic Advisor using the provided academic documents. The advisor should answer questions about courses, prerequisites, credits, regulations, semester offerings etc., while grounding its responses in the provided sources.
- The advisor should also be interactive: when a student's question does not contain enough information to provide a reliable answer, it should ask relevant follow-up questions rather than make assumptions.

Student: *“Can I take Machine Learning next semester?”*

AI: *“I can check that. What courses have you already completed, particularly the prerequisite courses for Machine Learning like EDA...?”*

Does asking the right questions actually improve the quality of the final recommendation?

Phase #2 — Challenge the Advisor

- Create a set of synthetic student profiles and challenging academic scenarios to test the advisor. Include cases involving missing information, ambiguous questions, insufficient prerequisites/credits, unavailable courses and conflicting academic rules.
- Evaluate whether the advisor can recognize when it does not have enough information, identify conflicts, and avoid making unsupported or incorrect recommendations.

Phase #3 — Improve Through Prompt Engineering

- Develop and test different prompting approaches to improve the advisor's performance. Compare a basic LLM/document-Q&A approach with increasingly structured approaches using techniques such as role/context, constraints, delimiters, output formatting, follow-up questioning and structured student information.
- Document which prompting techniques improve the quality, reliability and consistency of the advisor's responses.

Phase #4 — Evaluate and Measure

- Create a predefined test set of at least 20–30 academic queries/scenarios with verified expected answers or outcomes. Compare the different versions of your system quantitatively (e.g., percentage, count, or average response time) using:
 - Accuracy
 - Hallucination rate
 - Correct eligibility decisions
 - Source/evidence correctness
 - Response time
 - Handling of missing information
 - Handling of conflicting rules
 - Incorrect recommendations
- Explain your evaluation methodology and present the results clearly.

(Please note the distinction:

Correctness: Whether the answer/recommendation matches the verified academic rule or expected outcome.

Reliability: Whether the system consistently provides grounded answers, identifies uncertainty and avoids unsupported recommendations.

This is particularly useful because a system can be correct on easy questions but unreliable on ambiguous ones.)

For each test case, classify the response against a verified expected outcome/source and record whether it is correct, partially correct, unsupported/hallucinated or incorrect. The handling, evaluation methodology and scoring criteria must be clearly defined in the report.

Phase #5 — Compare and Analyze

- Analyze how the quality of the system changes across the development stages:

Basic LLM → Structured Prompting → RAG → RAG + Structured Student Data

- Handling Missing Information: Create at least 3 questions for which the provided data is insufficient to give a reliable answer. Does your system correctly acknowledge the missing information rather than inventing an answer? How did you handle it?
- Identify what improved, what did not improve and where the system still fails. Explain why RAG and structured information are useful compared with simply uploading documents and asking questions.

Phase #6 — Deploy and Demonstrate

- Deploy a working prototype of the AI Academic Advisor, prioritizing correctness, reliability and a seamless, clear, user-friendly interaction experience. Demonstrate the system using representative student scenarios and show how it handles normal, ambiguous, missing-information and conflicting-rule cases.

Phase #7 — Documentation & Presentation

- Submit a 10-page report, PPT presentation and working prototype documenting the system architecture, prompts, RAG approach, test scenarios, evaluation methodology, results, limitations and key findings.

5. Rubric for evaluation

Marks are awarded for the quality of the approach, experimentation, evaluation and analysis—not simply for using more sophisticated technology. A well-designed RAG system that demonstrates strong evidence and evaluation can score higher than a poorly designed but elaborate system.

Phase	Assessment Criteria	Marks
Phase 1	RAG implementation, quality of document retrieval, grounding of responses, use of provided academic sources, functional system	10
Phase 2	Quality and diversity of synthetic student profiles; challenging/edge-case scenarios; follow-up questioning; handling missing and conflicting information	5
Phase 3	Quality of prompt design; use of context, constraints, delimiters, output control, structured student information; evidence that prompting improves results	5
Phase 4	Quality of 20–30 test cases; clearly defined evaluation methodology; accuracy, hallucination, eligibility, source correctness, response time, incorrect recommendations, etc.	10
Phase 5	Meaningful comparison of Basic LLM → Prompting → RAG → RAG + Structured Data; quality of insights about improvements, failures and limitations	5
Phase 6	Working prototype, usability, clarity of interaction, appropriate follow-up questions, source/evidence presentation, handling of uncertainty	5
Phase 7	Quality of 10-page report and PPT; architecture, methodology, experiments, results, limitations and conclusions	5
Presentation	Quality presentation , answering all the questions	5
Total		50

Assignment #2

Assignment Title: “Can we make the AI Academic Advisor multimodal, secure, legally/ethically responsible, and capable of meaningful human-AI collaboration?”

Additional Data to be created:

- Synthetic Student Database — student profiles, completed/failed courses, credits, semester, minors, etc.
- Synthetic Academic Documents/Records — where required, create sample documents such as student transcripts, course registration records
- Synthetic Multimodal Content — sample images, audio or video inputs relevant to academic advising, where applicable.

Phases for Extension:

Phase #8 — Multimodal Extension

Extend the AI Academic Advisor beyond text by allowing it to work with **at least two modalities**—for example, generate an infographic explaining a course structure, create an audio explanation of an academic regulation or generate a short video/orientation clip for students. Explore additional features such as multimodal interaction, voice-based advising, autonomous planning, AI agents, while identifying which responsibilities should remain with humans. Document which AI models/tools have been used and evaluate the quality and limitations of each modality.

Phase #9 — Agentic Extension

- Extend the system into an agentic Academic Advisor in which separate components/agents perform tasks such as: Regulation Retrieval → Course Eligibility → Credit Calculation → Recommendation → Verification
- Compare the agentic version with your RAG-based system and discuss whether the additional complexity actually improves the results.

Phase #10 — Privacy & Security Audit

Conduct a **privacy and security audit** of the AI Academic Advisor. Identify what student/academic information could be sensitive, determine what information should or should not be provided to an AI system and test scenarios involving inappropriate data exposure, unauthorized requests, prompt injection and accidental disclosure. Propose safeguards for protecting student information.

Phase #11 — Synthetic Media, Copyright & Legal Analysis

Choose at least **three AI-generated outputs** from the project (text/image/audio/video) and investigate their **copyright, ownership, attribution, licensing and potential misuse**. Identify what rights or permissions are relevant when using university material, copyrighted content, student-created content or AI-generated content and document appropriate practices for your system.

Phase #12 — Deploy, Demonstrate, Document and Present

Deploy the final AI Academic Advisor as a working prototype and demonstrate its capabilities, limitations and key improvements through representative academic scenarios. Submit a comprehensive report and PPT documenting the system architecture, methodology, experiments, evaluation results, privacy/security considerations and key findings, followed by a live demonstration/presentation of the solution.

Rubric for evaluation

Marks are awarded based on the quality of the approach, experimentation, evaluation, critical analysis, responsible application of AI, and user-friendliness—not simply on the use of advanced technologies. A well-designed, intuitive and seamless multimodal solution with strong privacy, security, ethical and legal considerations can score higher than a technically complex solution that lacks meaningful evaluation, usability, or responsible AI practices.

Phase	Assessment Criteria	Marks
Phase #8 Multimodal Extension	Quality and meaningful integration of at least two modalities ; relevance to academic advising; quality of generated/processed outputs; appropriate model/tool selection; evaluation of strengths and limitations; user-friendly multimodal experience	12
Phase #9 Agentic Extension	Appropriate agent/workflow design; meaningful separation of tasks; integration between agents; comparison with the RAG-based system;	10

	evidence that agentic architecture provides value rather than unnecessary complexity	
Phase #10 Privacy & Security Audit	Identification of sensitive information and risks; testing of privacy/security scenarios; prompt-injection and unauthorized-access testing; quality of proposed safeguards; responsible handling of synthetic student data	10
Phase #11 Synthetic Media, Copyright & Legal Analysis	Analysis of at least 3 AI-generated outputs; understanding of copyright, ownership, attribution, licensing and potential misuse; identification of relevant legal/ethical considerations; practical recommendations for responsible use	8
Phase #12 Deploy, Demonstrate, Document & Present	Working and reliable prototype; seamless and user-friendly experience; quality of demonstration; system architecture and methodology; quality of report/PPT; clarity of results, limitations and conclusions	10
Total		50